Tuesday Review Document: Chapters 6–9

Prof. Jordan C. Hanson

Short Review Exercises

1. **Adders and Arithmetic Logic**

   A full-adder has inputs A, B, and C_{in}, and outputs Σ and C_{out}.

   (a) If $A = 1$, $B = 0$, and $C_{\text{in}} = 1$, determine Σ and C_{out}.

   (b) Write the Boolean expression for the sum output Σ in terms of A, B, and C_{in}.

   (c) Explain why several full-adders can be connected together to add two multi-bit binary numbers.

2. **Comparators, Decoders, and Encoders**

   (a) A 4-bit comparator receives

   $$A = 1010, \qquad B = 0111.$$

   Which comparator output is HIGH: $A > B$, $A = B$, or $A < B$?

   (b) A 4-to-16 decoder has input

   $$A_3A_2A_1A_0 = 1001.$$

   Which decimal output line should go HIGH?

   (c) A decimal-to-BCD encoder has the input line corresponding to decimal 6 HIGH. What 4-bit BCD code should appear at its output?

3. **Multiplexers and Demultiplexers**

   A 4-to-1 multiplexer has data inputs D_0 through D_3, select inputs S_1S_0, and output Y.

   (a) Which data input is connected to Y when

   $$S_1S_0 = 10?$$

   (b) If

   $$D_0 = 0, \quad D_1 = 1, \quad D_2 = 1, \quad D_3 = 0,$$

   what is Y when $S_1S_0 = 10$?

   (c) A 1-to-4 demultiplexer performs the reverse routing operation. If $D_{\text{in}} = 1$ and $S_1S_0 = 01$, which output receives the input bit?

4. **Latches and Flip-Flops**

   (a) An SR latch initially has $Q = 0$. If the SET input is asserted momentarily and then released, what is the new value of Q?

   (b) A D flip-flop has $D = 1$ immediately before its active clock edge. What value is stored at Q immediately after the edge?

   (c) A JK flip-flop has $J = 1$ and $K = 1$. If $Q = 1$ before the active clock edge, what is Q afterward?

5. **Shift Registers**

   Consider a 4-bit serial-in/serial-out shift register whose flip-flops are initially reset to 0.

   (a) A single 1 is shifted into the register, followed only by 0s. How many active clock edges are required before the 1 reaches the serial output?

   (b) If the clock frequency is 5 MHz, how much time passes before that bit reaches the output?

   (c) Briefly state the difference between *serial loading* and *parallel loading* of a register.

6. **Counters**

   (a) A 4-bit binary counter starts at 0000. Write the next four states after four active clock edges.

(b) A 4-bit asynchronous counter is driven by a 100 kHz clock. How many clock cycles are required to count from 0000 to 1111? How much time does this require?

(c) What is the main difference between a synchronous counter and an asynchronous (ripple) counter?

Design Problem

7. **Build a 4-bit Accumulator**

An **accumulator** is a digital system that repeatedly adds a new binary value to a previously stored result. You will design a 4-bit accumulator using components studied in class.

(a) Begin with a **4-bit register** whose output is

$$Q = Q_3Q_2Q_1Q_0.$$

Draw the register and label the four data inputs, the four outputs, the clock input `CLK`, and a reset input `RESET`. What value should be stored in the register immediately after `RESET` is activated?

(b) Add a **4-bit parallel adder** to the system. One 4-bit input to the adder is an external binary number

$$X = X_3X_2X_1X_0.$$

Connect the output of the register to the second 4-bit input of the adder. Draw these connections and label the four sum outputs

$$S = S_3S_2S_1S_0.$$

Write an arithmetic expression relating S, X, and the current register value Q.

(c) Connect the four sum outputs of the adder to the four data inputs of the register. Explain what happens on the next active clock edge.

Write an expression for the next register value:

$$Q_{\text{next}} =$$

(d) Complete the feedback path so that the system can repeatedly accumulate values. Explain why feeding the register output back into the adder causes the system to remember the sum of all previous inputs.

(e) Assume the accumulator is initially reset so that

$$Q = 0000.$$

The following values are presented to the input X, one value per clock cycle:

$$0011, \qquad 0010, \qquad 0100.$$

Determine the value stored in the register after each clock edge.

Clock	$\boldsymbol{X}$	$\boldsymbol{Q}$ **after edge**
0	RESET	0000
1	0011	
2	0010	
3	0100	

(f) Suppose one additional input is

$$X = 1000.$$

Determine the new 4-bit register value. What happens if the numerical result is larger than the largest value that can be stored in four bits?